

Co-VAE: Drug-Target Binding Affinity Prediction by Co-Regularized Variational Autoencoders

Tianjiao Li, Xing-Ming Zhao[✉], Senior Member, IEEE, and Limin Li[✉]

Abstract—Identifying drug-target interactions has been a key step in drug discovery. Many computational methods have been proposed to directly determine whether drugs and targets can interact or not. Drug-target binding affinity is another type of data which could show the strength of the binding interaction between a drug and a target. However, it is more challenging to predict drug-target binding affinity, and thus a very few studies follow this line. In our work, we propose a novel co-regularized variational autoencoders (Co-VAE) to predict drug-target binding affinity based on drug structures and target sequences. The Co-VAE model consists of two VAEs for generating drug SMILES strings and target sequences, respectively, and a co-regularization part for generating the binding affinities. We theoretically prove that the Co-VAE model is to maximize the lower bound of the joint likelihood of drug, protein and their affinity. The Co-VAE could predict drug-target affinity and generate new drugs which share similar targets with the input drugs. The experimental results on two datasets show that the Co-VAE could predict drug-target affinity better than existing affinity prediction methods such as DeepDTA and DeepAffinity, and could generate more new valid drugs than existing methods such as GAN and VAE.

Index Terms—Variational autoencoders, co-regularized VAE, drug-target binding affinity

1 INTRODUCTION

IN drug discovery and development, it is crucial to identify drug-target interactions (DTI), for example, to identify a new target which selectively binds to an existing drug, or identify a new drug which binds to a particular target. The approved DTIs, the heterogeneous chemical or structural profilings for drugs and targets are often used to predict new DTIs. Another type of related data which gains more interest in recent years is the drug-target binding affinity value, which is usually measured by dissociation constant (K_d), inhibition constant (K_i) or the half maximal inhibitory concentration (IC_{50}). The drug-target binding affinity shows the strength of the protein-ligand interaction between a drug and a target. It provides more information for drug-target interactions, and thus it is important to predict the affinity values for studying DTIs.

The existing approaches for studying drug-target interactions can be divided to two classes: predicting the binary DTIs and predicting the binding affinity values. The first class is to predict DTIs directly, i.e., to determine whether a drug and a target interact or not, by known drug-target interactions and

their feature vectors or pairwise similarities extracted by combining chemical or structural descriptors of drugs with descriptors for targets [1], [2], [3]. Most methods formulate the prediction of DTIs as a binary classification problem [4], [5], [6], [7], a clustering problem [8], or a matrix completion problem, based on either feature vectors of the drugs or the targets, or pairwise similarities among the drugs or the targets. For example, a drug-target pair can be considered as positive if they are known to have interaction, and negative otherwise. Classification models such as Support Vector Machines (SVM) or Neural Networks (NN) can be learnt by training on the positive and negative drug-target pairs whose features or pairwise similarities can be constructed by the chemical or structural profilings of drugs or targets. The basic idea for the clustering approaches is to first project the drugs and targets based on their profilings and the approved DTIs to a common hidden space and then to co-cluster the drugs and targets in this hidden space. The drugs and targets in the same cluster are considered to have interactions.

The above direct methods to predict DTIs have received success since last decade. However, these methods have two major limitations. One is that the true negative drug-target pairs are unknown, and the pairs with missing values are often taken as pseudo negative pairs in the training process. The other is that the binary labels for the drug-target pairs could not provide more information about how strongly a drug binds to a target.

To alleviate the above limitations of direct methods, the other line for studying drug-target interactions is to predict drug-target binding affinity. Three types of affinity values are commonly used in drug discovery. One type of affinity value is dissociation constant (K_d), which reflects the affinity of the ligands for the target. Another type of affinity value is the inhibition constant (K_i), which indicates the inhibitory intensity of the target by the inhibitor. Half maximal inhibitory concentration (IC_{50}) indicates the concentration of a drug or

- Tianjiao Li and Limin Li are with the School of Mathematics and Statistics, Xi'an Jiaotong University, Xi'an 710049, China. E-mail: litj468@stu.xjtu.edu.cn, liminli@mail.xjtu.edu.cn.
- Xing-Ming Zhao is with the Institute of Science and Technology for Brain-Inspired Intelligence, Fudan University, China, and also with the MOE Key Laboratory of Computational Neuroscience and Brain-Inspired Intelligence, and MOE Frontiers Center for Brain Science, Shanghai 200433, China. E-mail: xmzhao@fudan.edu.cn.

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(Corresponding authors: Xing-Ming Zhao and Limin Li.)

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inhibitor required when a biological process is inhibited by half. There are several techniques available for measuring binding affinity, such as ELISA, gel-shift analysis, pull down analysis, equilibrium dialysis, analytical ultracentrifugation, SPR, and spectroscopic analysis. For example, affinities are detected by the change of the dielectric constant caused by molecular adsorption on the heavy metal film in SPR. In the SPR, the antibody is bound to the enzyme complex, and then the affinity is detected by color development.

Since the binding affinity is measured continuously, the second class of methods often formulate the affinity prediction problem as a regression problem. Similarity-based methods and feature-based methods have been both proposed to predict binding affinity. The work in [9] proposes a read-across method *KronRLS* based on similarities among drug-target pairs defined through the Kronecker product of a drug-drug similarity matrix and a target-target similarity matrix. One drawback of this method is that it can only capture linear dependencies in the training data. The study in [10] proposes a non-linear method, *SimBoost*, which constructs features for drugs, targets, and drug-target pairs based on the similarity between the drugs and the similarity between the targets. Gradient boosting machines are then used in *SimBoost* to predict affinity of a drug-target pair and to predict an affinity interval. Both the *KronRLS* and *SimBoost* methods take the drug-drug similarities and target-target similarities as input, and they may ignore the features hidden in the original drug and target structures. The work in [11] proposes a deep learning method *DeepDTA* which takes the structures of drugs and targets as input to learn more informative features for predicting drug-target binding affinity. *DeepAffinity* [12] computes deep features for drug sequences and protein sequences by pretrained unsupervised Seq2Seq autoencoder models [37], then learn a regression model for predicting affinity. There are also methods such as *GraphDTA* [13] which represent drugs as graphs and make use of graph neural network to predict drug-target affinity. One major limitation of the method is that it is designed for predicting the missing affinity values between old drugs and old targets. When a new drug or new target is given, it is still challenging to predict the affinity values between the new drug and the old targets, or the values between the new target and the old drugs. Furthermore, it is lack of theoretical support for these methods.

Recently, variational autoencoders (VAE) [14] has become a popular model in the field of machine learning to learn the latent feature z for the input variable x and thus reconstruct x as output. Different with simple auto-encoder, the VAE can learn the distribution of hidden feature vectors, and generate a new sample from the distribution. The VAE model is a generative model and with good theoretical foundation. The VAE has been successfully applied in image processing [15], [16] and text processing [17]. Although the VAE is powerful in many fields of machine learning, it could not be used for predicting drug-target affinity directly. Our contributions of this work are as follows.

1. We propose a novel co-regularized variational autoencoder (Co-VAE) model which generalizes the VAE model to predict drug-target affinities for new drugs or new targets. Our Co-VAE model consists of

two VAEs for drugs and targets, respectively, and a co-regularized block.

2. The Co-VAE model could generate new drugs (or targets) which share similar targets (drugs) for given drugs (targets), respectively.
3. We theoretically show that the Co-VAE is to maximize the lower bound of log likelihood of the joint distribution of triplets (drug, target, affinity) under certain assumptions.

Our Co-VAE model takes use of the VAE in the deep learning architecture for drug-target affinity prediction. It first extracts high-level features for drugs or targets from drug SMILES structures or target sequences via CNN blocks in probabilistic encoder, and then reconstructs drug structures or target sequences in probabilistic decoder. The obtained features for drugs and targets are combined into a co-regularized block whose loss is defined by the affinity errors between the ground truth values and the predicted values. We also take the advantage of the Gated convolutional neural networks (GatedCNN), which is modeled on a threshold mechanism in LSTM [19] with a multi-layer CNN structure. The advantage of our Co-VAE model is that it could capture high-level features for reconstructing the drugs and targets, based on which the model could not only predict the affinity values between new drug and new target, but also generate new drug which shares similar targets with the input drug. Our experiments on two continuous binding affinities datasets, Davis dataset [20] and KIBA dataset [10], [21] show the advantages of our Co-VAE model over the existing methods including *KronRLS* [9], *DeepDTA* [11], *DeepAffinity* [12] and *GraphDTA* [13]. We also evaluated the generative performance of the Co-VAE model on KIBA dataset, and the results show that the Co-VAE model could generate more new valid drugs than other generative methods such as GAN and VAE.

1.1 Related Work

1.1.1 *KronRLS* [9]

Regularized Least Squares Models (RLS) is a feasible method of predicting binary drug-target interactions [26]. *KronRLS* [9] is a generalized RLS model for the prediction of continuous binding affinities. Given a set X of drug-target pairs $x_i = (d_i, t_i)$ and a vector y consisting of their binding affinities y_i , $i = 1, \dots, m$, *KronRLS* learns prediction function f by minimizing the following loss:

$$\mathcal{L}(f) = \sum_{i=1}^m (y_i - f(x_i))^2 + \lambda \|f\|_k^2, \quad (1)$$

where $\lambda > 0$ is a regularization parameter balancing the prediction error and model complexity, and $\|f\|_k^2$ is the norm of f measured in the Hilbert space associated with a kernel function k . According to the theorem in [27], a minimizer of the Equation (1) can be expressed as follows:

$$f(x) = \sum_{i=1}^m a_i k(x, x_i), \quad (2)$$

where the kernel function k is a similarity measure between two drug-target pairs. All kernels between $x_i, i = 1, \dots, m$ can be represented by an $m \times m$ matrix K which can be

computed as $K = K_d \otimes K_t$, where K_d and K_t , drug and target kernels, come from similarity matrix based on drug chemical structure and target sequence, $\otimes$ denotes Kronecker product. The parameter a_i in Equation (2) can be found via solving the following linear equations:

$$(\mathbf{K} + \lambda \mathbf{I})\mathbf{a} = \mathbf{y},$$

where a_i is element of the $\mathbf{a}$, $\mathbf{I}$ is the $m * m$ identity matrix. Besides, to avoid the computation of the large Kronecker product matrix, KronRLS suggests to use Kronecker algebraic optimization [28].

1.1.2 DeepDTA [11]

DeepDTA is a deep-learning based model which uses sequence information of drug and target to predict binding affinities instead of drug and target similarity matrix [11]. The proposed methodology converts drug and target sequences into matrix as network input, then uses two Convolutional Neural Network (CNN) blocks to extract drug and target features. Each CNN block has three 1D-Convolutional layers. After that drug and target features are fed into another block which is composed of several fully connected (FC) layers and combined into affinity. Since DeepDTA treated drug-target binding affinities prediction as a regression problem, the loss function of the model is mean squared error between predictive affinities and true affinities

$$\mathcal{L}_{\text{DeepDTA}} = \frac{1}{n} \sum_{i=1}^n (P_i - Y_i)^2,$$

where P is the network prediction vector, Y is the true affinities vector corresponding to P , n indicates the number of samples.

1.1.3 DeepAffinity [12]

The DeepAffinity method also uses the sequence information of the drug and target to predict drug-target affinity. A recurrent neural network (RNN) model, Seq2Seq [37], is used in DeepAffinity, was initially used in natural language processing and then applied to embedding compound SMILES strings into fingerprints [38]. The Seq2Seq model is an auto-encoder that consists of two recurrent units known as the encoder and the decoder, respectively. The encoder maps an input SMILES sequence to a fixed-dimension vector, and the decoder maps this vector to the target SMILES sequence. Deepaffinity simply computes deep features for drugs or proteins by a pretrained unsupervised Seq2Seq autoencoder model, then learn a regression model for predicting affinity.

1.1.4 GraphDTA [13]

GraphDTA uses sequence representations for targets and graph representations for drugs. The GraphDTA model first makes use of the Rdkit toolkit to convert the drug SMILES sequence into a molecular graph, and then uses the graph neural network GCN to extract the features of the drug. After that, the CNN network is used to extract the features for the target sequence, and finally several fully connected layers are also used to obtain affinity.

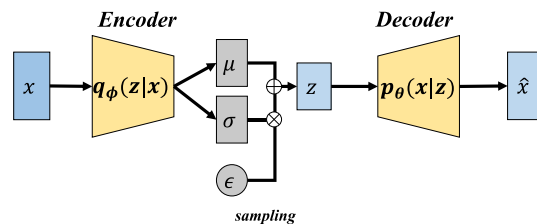


Fig. 1. The architecture of VAE model. VAE contains two parts, an encoder and a decoder. The encoder is used to obtain the variable features. In general, the reparameterization trick is used. Decoder is used to transform features into original variable structures. In this figure, $\otimes$ means element-wise product.

2 METHODS

2.1 Problem Definition

Suppose we are given a set of drugs D , where each drug is represented by its SMILES string, and a set of protein targets T , where each target is represented by its amino-acid sequences. The affinity matrix I contains the affinity values among the above drugs and targets. For a given new drug d or a new target t , our goal in this work is to predict the affinity values between the drug d and the targets in T , or the values between the target t and the drugs in D .

2.2 Variational Autoencoder

The variational autoencoder (VAE, [14]) is to learn a distribution for multi-dimensional variable $\mathbf{x}$ through a latent variable $\mathbf{z}$ based on N i.i.d. samples $\mathbf{X} = \{\mathbf{x}^i\}_{i=1}^N$ of $\mathbf{x}$. It combines techniques from deep learning and Bayesian machine learning, specifically variational inference.

The framework of the VAE model can be divided into two parts (shown in Fig. 1): a recognition model $q_\phi(\mathbf{z}|\mathbf{x})$ (probabilistic encoder) which produces the distribution over the latent variable $\mathbf{z}$ from a given $\mathbf{x}$, and a generative model $p_\theta(\mathbf{x}|\mathbf{z})$ (probabilistic decoder) which produces the distribution over $\mathbf{x}$ from a given $\mathbf{z}$. Note that the recognition model $q_\phi(\mathbf{z}|\mathbf{x})$ is an approximation to the intractable true posterior $p_\theta(\mathbf{z}|\mathbf{x})$. The VAE learns the recognition model parameters ϕ jointly with the generative model parameters θ by maximizing the following lower bound of the marginal likelihood $\log p_\theta(\mathbf{x}^i)$

$$\mathcal{L}(\theta, \Phi; \mathbf{x}^i) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x}^i)} \log p_\theta(\mathbf{x}^i|\mathbf{z}) - D_{KL}(q_\phi(\mathbf{z}|\mathbf{x}^i) || p_\theta(\mathbf{z})), \quad (3)$$

where $p_\theta(\mathbf{z})$ is the prior for $\mathbf{z}$, $D_{KL}(p||q) = -\int_{\mathcal{X}} p(\mathbf{x}) \log \frac{p(\mathbf{x})}{q(\mathbf{x})} d\mathbf{x}$ is the KL Divergence for two distributions p and q . The objective function of the VAE model is the combination of two terms. The first term is the expected log likelihood and is responsible for the reconstruction penalty, while the second term is the KL-divergence and is responsible for the regularization penalty.

The VAE takes the prior over the latent variable $\mathbf{z}$ as $p_\theta(\mathbf{z}) = \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})$, and assumes the variational distribution $q_\phi(\mathbf{z}|\mathbf{x})$ is multivariate Gaussian

$$\log q_\phi(\mathbf{z}|\mathbf{x}^i) = \log \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}^i, \boldsymbol{\sigma}^i \mathbf{I}),$$

where $\boldsymbol{\mu}^i$ and $\boldsymbol{\sigma}^i$ are outputs of encoder for $\mathbf{x}^i$. The VAE further uses the reparameterization trick $\mathbf{z}^{i,l} = \boldsymbol{\mu}^i + \boldsymbol{\sigma}^i \odot \boldsymbol{\epsilon}^l$, $\boldsymbol{\epsilon}^l \sim$

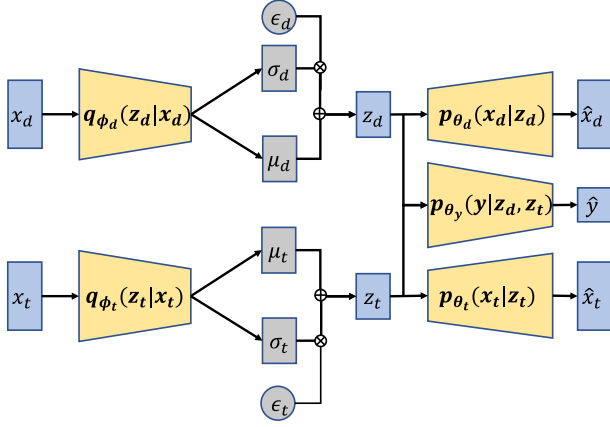


Fig. 2. The graphical structure for Co-VAE model.

$\mathcal{N}(\mathbf{0}, \mathbf{I})$, where $\odot$ means element-wise product, to sample from the posterior $q_{\Phi}(\mathbf{z}|\mathbf{x}^i)$. By putting these assumptions to the Equation (3), one can obtain

$$\begin{aligned} \mathcal{L}(\theta, \Phi; \mathbf{x}^i) &\approx \frac{1}{L} \sum_{l=1}^L (\log p_{\theta}(\mathbf{x}^i | \mathbf{z}^{i,l})) - D_{KL}(q_{\Phi}(\mathbf{z} | \mathbf{x}^i) || p_{\theta}(\mathbf{z})) \\ &= \frac{1}{L} \sum_{l=1}^L (\log p_{\theta}(\mathbf{x}^i | \mathbf{z}^{i,l})) \\ &\quad + \frac{1}{2} \sum_{j=1}^J (1 + \log((\sigma_j^i)^2) - (\mu_j^i)^2 - (\sigma_j^i)^2), \end{aligned} \quad (4)$$

where J is the dimension for the latent variable $\mathbf{z}$, σ_j^i and μ_j^i are the j th element of σ and μ , respectively, and $p_{\theta}(\mathbf{x}^i | \mathbf{z}^{i,l})$ is a Bernoulli or Gaussian multi-layered perceptron.

Neural networks are used in VAE as probabilistic encoders and decoders. The encoder network takes the training data $\mathbf{x}^i$ and predicts the parameters (mean μ^i and covariance σ^i) of the variational distribution $q_{\Phi}(\mathbf{z}|\mathbf{x}^i)$. By generating a sample $\mathbf{z}^i$ from this distribution as a input, the decoder network makes a prediction for the data based on this sample. The objective function of the VAE in Equation (4) hopes that output data examples should have high probability and the variational distribution should be similar to the prior.

2.3 The Co-VAE Model

The VAE is a state-of-the-art for generative models to find out the marginal distribution $p(\mathbf{x})$ for a high-dimensional random vector $\mathbf{x}$. However, in our scenario, it could not be used directly to predict drug-target binding affinity. In this section, we propose a Co-VAE model which could not only predict affinities but also generate new drugs or targets.

We first formulate our drug-target affinity prediction problem as follows. Let $\mathbf{x}_d$ and $\mathbf{x}_t$ be the random vectors for a drug and a target, and y be the random variable for the affinity values. We aim to learn a predictive model $p(y|\mathbf{x}_d, \mathbf{x}_t)$ based on the training triplets $\{\mathbf{x}_d^i, \mathbf{x}_t^i, y^i\}_{i=1}^N$. We assume the latent variables for drug $\mathbf{x}_d$ and target $\mathbf{x}_t$ as $\mathbf{z}_d$ and $\mathbf{z}_t$, respectively.

We propose a Co-VAE model as shown in Fig. 2. The Co-VAE model contains two recognition models $q_{\phi_d}(\mathbf{z}_d|\mathbf{x}_d)$ and

$q_{\phi_t}(\mathbf{z}_t|\mathbf{x}_t)$, which independently produce the latent variables $\mathbf{z}_d$ and $\mathbf{z}_t$ from the input $\mathbf{x}_d$ and $\mathbf{x}_t$, respectively, and three generative models $p_{\theta_d}(\mathbf{x}_d|\mathbf{z}_d)$, $p_{\theta_t}(\mathbf{x}_t|\mathbf{z}_t)$ and $p_{\theta_y}(y|\mathbf{z}_d, \mathbf{z}_t)$, which independently produce the distributions over $\mathbf{x}_d$, $\mathbf{x}_t$ and y , respectively, from $\mathbf{z}_d$ and $\mathbf{z}_t$. Based on the assumption of the Co-VAE model, we have the following theorem for the lower bound of log likelihood of the joint distribution of $(\mathbf{x}_d^i, \mathbf{x}_t^i, y^i)$, which is taken as the objective function of the Co-VAE model.

Theorem 2.1. *With the assumption of the independence between the priors $p_{\theta_d}(\mathbf{z}_d)$ and $p_{\theta_t}(\mathbf{z}_t)$, the graphical model shown in Fig. 2 generates a lower bound of log likelihood of the joint distribution of $(\mathbf{x}_d^i, \mathbf{x}_t^i, y^i)$ as*

$$\begin{aligned} \log p_{\theta}(\mathbf{x}_d^i, \mathbf{x}_t^i, y^i) &\geq \mathcal{L}(\theta, \Phi; \mathbf{x}_d^i, \mathbf{x}_t^i, y^i) = \\ &\underbrace{\mathbb{E}_{q_{\phi_d}(\mathbf{z}_d|\mathbf{x}_d^i)} \log p_{\theta_d}(\mathbf{x}_d^i|\mathbf{z}_d) - D_{KL}(q_{\phi_d}(\mathbf{z}_d|\mathbf{x}_d^i) || p_{\theta_d}(\mathbf{z}_d))}_{\mathcal{L}_{\text{DrugVAE}}} \\ &\quad + \underbrace{\mathbb{E}_{q_{\phi_t}(\mathbf{z}_t|\mathbf{x}_t^i)} \log p_{\theta_t}(\mathbf{x}_t^i|\mathbf{z}_t) - D_{KL}(q_{\phi_t}(\mathbf{z}_t|\mathbf{x}_t^i) || p_{\theta_t}(\mathbf{z}_t))}_{\mathcal{L}_{\text{TargetVAE}}} \\ &\quad + \underbrace{\mathbb{E}_{q_{\Phi}(\mathbf{z}_d, \mathbf{z}_t|\mathbf{x}_d^i, \mathbf{x}_t^i)} \log p_{\theta_y}(y^i|\mathbf{z}_d, \mathbf{z}_t)}_{\mathcal{L}_{\text{CoREG}}}, \end{aligned} \quad (5)$$

where $\theta = \{\theta_d, \theta_t, \theta_y\}$ and $\Phi = \{\phi_d, \phi_t\}$.

Proof. Based on the lower bound of likelihood shown in the inequality (3), the log likelihood of the joint distribution of $(\mathbf{x}_d^i, \mathbf{x}_t^i, y^i)$ has the following lower bound:

$$\begin{aligned} \log p_{\theta}(\mathbf{x}_d^i, \mathbf{x}_t^i, y^i) &\geq \mathcal{L}(\theta, \Phi; \mathbf{x}_d^i, \mathbf{x}_t^i, y^i) \\ &= \mathbb{E}_{q_{\Phi}(\mathbf{z}_d, \mathbf{z}_t|\mathbf{x}_d^i, \mathbf{x}_t^i, y^i)} \log p_{\theta}(\mathbf{x}_d^i, \mathbf{x}_t^i, y^i | \mathbf{z}_d, \mathbf{z}_t) \\ &\quad - D_{KL}(q_{\Phi}(\mathbf{z}_d, \mathbf{z}_t | \mathbf{x}_d^i, \mathbf{x}_t^i, y^i) || p_{\theta}(\mathbf{z}_d, \mathbf{z}_t)), \end{aligned} \quad (6)$$

where $q_{\Phi}(\mathbf{z}_d, \mathbf{z}_t | \mathbf{x}_d^i, \mathbf{x}_t^i, y^i)$ is an approximation of the posterior $p_{\theta}(\mathbf{z}_d, \mathbf{z}_t | \mathbf{x}_d^i, \mathbf{x}_t^i, y^i)$.

By the graphical model as shown in Fig. 2, we have

$$p_{\theta}(\mathbf{x}_d^i, \mathbf{x}_t^i, y^i | \mathbf{z}_d, \mathbf{z}_t) = p_{\theta_d}(\mathbf{x}_d^i | \mathbf{z}_d) p_{\theta_t}(\mathbf{x}_t^i | \mathbf{z}_t) p_{\theta_y}(y^i | \mathbf{z}_d, \mathbf{z}_t).$$

Because

$$q_{\Phi}(\mathbf{z}_d, \mathbf{z}_t | \mathbf{x}_d^i, \mathbf{x}_t^i, y^i) = q_{\phi_d}(\mathbf{z}_d | \mathbf{x}_d^i) q_{\phi_t}(\mathbf{z}_t | \mathbf{x}_t^i),$$

the first term in (6) can be simplified as

$$\begin{aligned} &\mathbb{E}_{q_{\phi_d}(\mathbf{z}_d|\mathbf{x}_d^i)} \log p_{\theta_d}(\mathbf{x}_d^i|\mathbf{z}_d) + \mathbb{E}_{q_{\phi_t}(\mathbf{z}_t|\mathbf{x}_t^i)} \log p_{\theta_t}(\mathbf{x}_t^i|\mathbf{z}_t) \\ &\quad + \mathbb{E}_{q_{\Phi}(\mathbf{z}_d, \mathbf{z}_t|\mathbf{x}_d^i, \mathbf{x}_t^i)} \log p_{\theta_y}(y^i|\mathbf{z}_d, \mathbf{z}_t). \end{aligned} \quad (7)$$

Due to the independence between the priors $p_{\theta_d}(\mathbf{z}_d)$ and $p_{\theta_t}(\mathbf{z}_t)$, we have $p_{\theta}(\mathbf{z}_d, \mathbf{z}_t) = p_{\theta_d}(\mathbf{z}_d) p_{\theta_t}(\mathbf{z}_t)$. Thus the second term in (6) is

$$D_{KL}(q_{\phi_d}(\mathbf{z}_d|\mathbf{x}_d^i) || p_{\theta_d}(\mathbf{z}_d)) + D_{KL}(q_{\phi_t}(\mathbf{z}_t|\mathbf{x}_t^i) || p_{\theta_t}(\mathbf{z}_t)). \quad (8)$$

By combining the Equations (7) and (8), the lower bound of likelihood of the joint distribution in (5) can be obtained. $\square$

For each drug-target pair $(\mathbf{x}_d^i, \mathbf{x}_t^i)$, our Co-VAE model is thus to maximize

$$\mathcal{L} = \mathcal{L}_{\text{DrugVAE}} + \mathcal{L}_{\text{TargetVAE}} + \mathcal{L}_{\text{CoREG}},$$

where $\mathcal{L}_{\text{DrugVAE}}$ and $\mathcal{L}_{\text{TargetVAE}}$ represent the VAE bound for drugs and targets, respectively, and $\mathcal{L}_{\text{CoREG}}$ is a co-regularized term which represents the regression bound responsible for the affinity reconstruction penalty for a pair of drug and target. Theorem 2.1 shows that the Co-VAE model is to maximize the lower bound of the joint distribution of triplet (drug, target, affinity).

For $\mathcal{L}_{\text{DrugVAE}}$ and $\mathcal{L}_{\text{TargetVAE}}$, let $p_{\theta_d}(\mathbf{x}_d|\mathbf{z}_d)$ and $p_{\theta_t}(\mathbf{x}_t|\mathbf{z}_t)$ be categorical distributions, whose probabilities are computed from $\mathbf{z}_d$ and $\mathbf{z}_t$ through the neural network, and thus we have

$$\begin{aligned}\log p_{\theta_d}(\mathbf{x}_d^i|\mathbf{z}_d) &= \mathbf{x}_d^{iT} \log \hat{\mathbf{x}}_d^i \\ \log p_{\theta_t}(\mathbf{x}_t^i|\mathbf{z}_t) &= \mathbf{x}_t^{iT} \log \hat{\mathbf{x}}_t^i,\end{aligned}$$

where $\hat{\mathbf{x}}_d^i = \mathbf{f}_{\theta_d}(\mathbf{z}_d)$ and $\hat{\mathbf{x}}_t^i = \mathbf{f}_{\theta_t}(\mathbf{z}_t)$ are the reconstructed drug and target, respectively. For $\mathcal{L}_{\text{CoREG}}$, we further assume $p_{\theta_y}(y|\mathbf{z}_d, \mathbf{z}_t)$ to be a normal distribution, whose probability are computed from $\mathbf{z}_d$ and $\mathbf{z}_t$ through the network, then we have

$$\log p_{\theta_y}(y^i|\mathbf{z}_d, \mathbf{z}_t) = -\lambda(y^i - \hat{y}^i)^2 + C, \quad (9)$$

where $\hat{y}^i = f_{\theta_y}(\mathbf{z}_d, \mathbf{z}_t)$, C is a constant, λ is the co-regularization parameter.

The Co-VAE takes the prior over the latent variables $\mathbf{z}_d$ and $\mathbf{z}_t$ as $p_{\theta_d}(\mathbf{z}_d) = \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $p_{\theta_t}(\mathbf{z}_t) = \mathcal{N}(\mathbf{0}, \mathbf{I})$, respectively, and assumes the variational distribution $q_{\phi_d}(\mathbf{z}_d|\mathbf{x}_d^i)$ to be multivariate Gaussians

$$\begin{aligned}\log q_{\phi_d}(\mathbf{z}_d|\mathbf{x}_d^i) &= \log \mathcal{N}(\boldsymbol{\mu}_d^i, \boldsymbol{\sigma}_d^{i2} \mathbf{I}), \\ \log q_{\phi_t}(\mathbf{z}_t|\mathbf{x}_t^i) &= \log \mathcal{N}(\boldsymbol{\mu}_t^i, \boldsymbol{\sigma}_t^{i2} \mathbf{I}),\end{aligned}$$

where $\boldsymbol{\mu}_d^i$ and $\boldsymbol{\sigma}_d^i$ are outputs of encoder for $\mathbf{x}_d^i$, and $\boldsymbol{\mu}_t^i$ and $\boldsymbol{\sigma}_t^i$ are outputs of encoder for $\mathbf{x}_t^i$. By using the reparameterization trick $\mathbf{z}_d^{i,l} = \boldsymbol{\mu}_d^i + \boldsymbol{\sigma}_d^i \odot \boldsymbol{\epsilon}_d^{i,l}$, $\boldsymbol{\epsilon}_d^{i,l} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ to sample from the posterior $q_{\phi_d}(\mathbf{z}_d|\mathbf{x}_d^i)$, and $\mathbf{z}_t^{i,l} = \boldsymbol{\mu}_t^i + \boldsymbol{\sigma}_t^i \odot \boldsymbol{\epsilon}_t^{i,l}$, $\boldsymbol{\epsilon}_t^{i,l} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ to sample from the posterior $q_{\phi_t}(\mathbf{z}_t|\mathbf{x}_t^i)$, the objective function to maximize in the Co-VAE model at point $(\mathbf{x}_d^i, \mathbf{x}_t^i, y^i)$ is

$$\begin{aligned}\mathcal{L}(\theta, \phi; \mathbf{x}_d^i, \mathbf{x}_t^i, y^i) &\approx \frac{1}{L_d} \sum_{l=1}^{L_d} \mathbf{x}_d^{iT} \log \mathbf{f}_{\theta_d}(\mathbf{z}_d^{i,l}) \\ &+ \frac{1}{2} \sum_{j=1}^{J_d} (1 + \log((\sigma_{d,j}^i)^2) - (\mu_{d,j}^i)^2 - (\sigma_{d,j}^i)^2) \\ &\quad + \frac{1}{L_t} \sum_{l=1}^{L_t} \mathbf{x}_t^{iT} \log \mathbf{f}_{\theta_t}(\mathbf{z}_t^{i,l}) \\ &+ \frac{1}{2} \sum_{j=1}^{J_t} (1 + \log((\sigma_{t,j}^i)^2) - (\mu_{t,j}^i)^2 - (\sigma_{t,j}^i)^2) \\ &\quad - \lambda \sum_{l_d=1}^{L_d} \sum_{l_t=1}^{L_t} (y^i - f_{\theta_y}(\mathbf{z}_d^{i,l_d}, \mathbf{z}_t^{i,l_t}))^2,\end{aligned} \quad (10)$$

where $\mu_{d,j}^i$, $\mu_{t,j}^i$, $\sigma_{d,j}^i$ and $\sigma_{t,j}^i$ represent the j th element in the vectors $\boldsymbol{\mu}_d^i$, $\boldsymbol{\mu}_t^i$, $\boldsymbol{\sigma}_d^i$ and $\boldsymbol{\sigma}_t^i$, respectively, J_d and J_t are the dimensions for the latent variables $\mathbf{z}_d$ and $\mathbf{z}_t$, respectively.

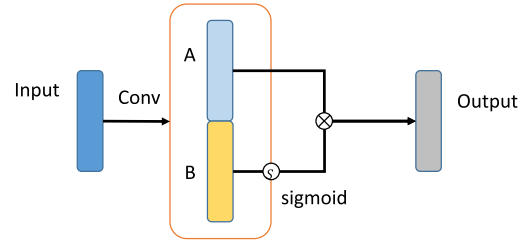


Fig. 3. The architecture of the GatedCNN. The input is divided into two units after convolution, unit A and unit B. Unit A is multiplied with sigmoid B to obtain the output.

2.4 GatedCNN

To predict drug-target affinity, neural networks such as Convolutional Neural Network (CNN,[22]) can be used to extract high-level features for drug and target strings. Since a drug is represented as a SMILE string, and a target protein is represented as an amino-acid sequence, language processing models can be used to extract better features for drugs and targets.

In our Co-VAE model for drug-target affinity prediction, we use gated convolutional networks (GatedCNN) [18], as shown in Fig. 3. The GatedCNN has been shown to outperform other language model such as LSTMs [19] in many real applications. Similar to LSTMs, GatedCNN adds a gate after every convolutional layer to control the information passed in the flow. The input of GatedCNN layer is word embedding matrix. Different from ordinary convolution, the output of gated convolution is divided into two parts. One part is the convolution activation value, denoted by unit A, and the other part is the gate value, denoted as unit B, which functions as a gate unit to control the output of A. The final output is the multiplication of A and sigmoid B. The convolution unit A and the gate unit B form a gated convolution layer. GatedCNN has excellent performance in many aspects, such as extracting better features, speeding up the training process, greatly shortening the training time, and avoiding problems such as gradient disappearance.

2.5 Network Structure in Co-VAE

The network structure of our Co-VAE model has five parts: drug encoder, target encoder, drug decoder, target decoder, co-regularized block, as shown in Fig. 4.

Drug encoder and target encoder both consist of label encoding, embedding layer and GatedCNN blocks. Label encoding layer simply transforms each character in the input string to an integer based on an established character dictionary, and the embedding layer converts each integer into a vector with the same size. Thus the input sequence is finally converted into an embedding matrix after the embedding layer. In our model, the architecture of GatedCNN block has three 1D-gated CNN layers, two Rectified Linear Unit (ReLU) [23] activation functions and a max-pooling layer. There is a ReLU activation function after each 1D-convolutional layer. The numbers of filters in the second layer and third layer are twice and three times that of the first layer, respectively. The max-pooling layer is at the end of encoder to compress the extracted features.

In drug decoder and target decoder, deconvolutional networks [24] are used to reconstruct the original inputs based on the features and feature filters. Each decoder consists of

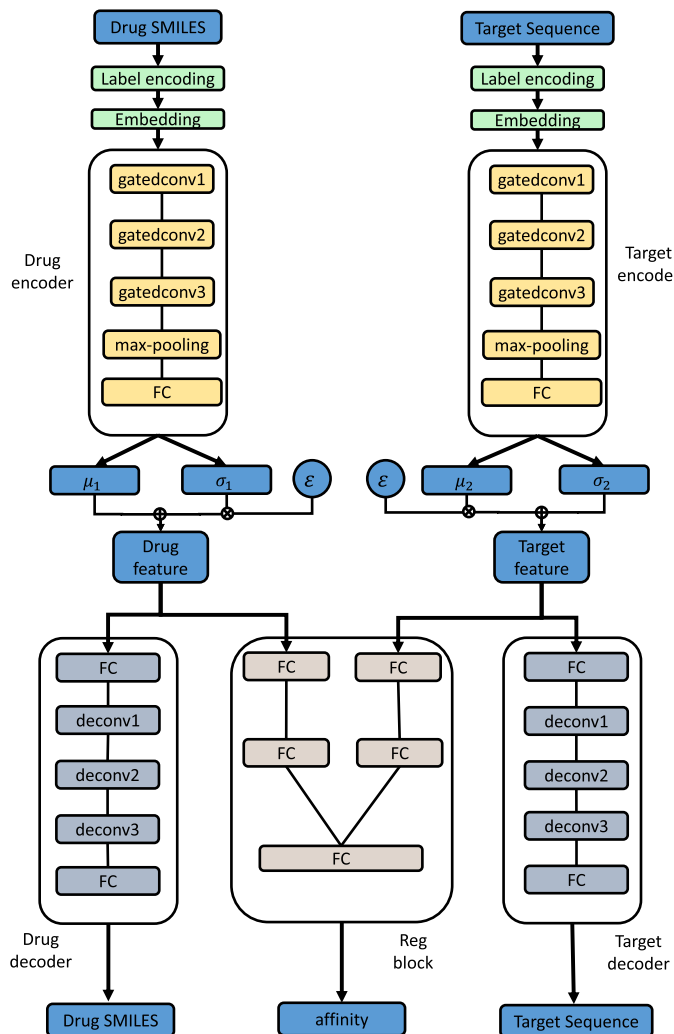


Fig. 4. The structure diagram for the Co-VAE model. The Co-VAE is divided into five parts: drug encoder, target encoder, drug decoder, target decoder and reg block. The drug encoder and target encoder are used to extract the features of drug SMILES and target sequences, respectively, the drug decoder and target decoder are used to reconstruct drug SMILES and target sequences, and the Reg block obtains the affinity value based on the features of the input drug SMILES and target sequence.

a FC layer, three deconvolutional layers and another FC layer. The first FC layer converts drug or target features into new features with right size for deconvolutional layers. Then three deconvolutional layers reconstruct drug or target from new features. The final FC layer transforms deconvolutional layers output into drug or target which has the same size as the drug input or target input. After decoding by the decoder, we can get new drug SMILES string and new target sequence.

In Reg block, we first use two FC layers for the features of drug and target, and then connect them by another FC layer. There is an activation function after each FC layer, and a dropout layer between every two FC layers. Dropout is a trick in training deep neural networks. In each training batch, over-fitting can be significantly reduced by ignoring some feature detectors (making the hidden layer nodes have a value of 0) [25]. The final FC layer outputs a scalar which is a predicted affinity corresponding to the drug-target pair.

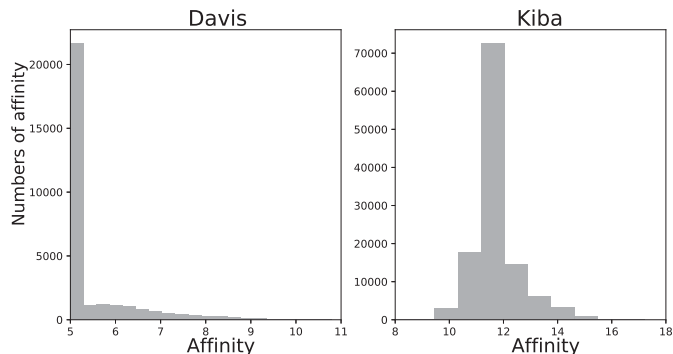


Fig. 5. The frequency histogram of the affinities in Davis and KIBA dataset.

3 EXPERIMENTS AND RESULTS

3.1 Datasets

We evaluated the performance of the proposed Co-VAE method on two drug–target affinity datasets, Davis dataset [20] and KIBA dataset [21], which were used in [11] previously. The drug SMILES strings are extracted from Pubchem database according to their PubChem IDs and the target sequences are from the UniProt protein database based on gene names or RefSeq accession numbers.

Davis dataset Davis dataset consists of 68 drugs and 442 targets, and the pairwise affinities measured by K_d value (kinase dissociation constant). The K_d affinity value for a drug and target reflects their binding ability. The range of affinities is $[0.016, 10000]$. The affinities in Davis dataset are K_d , which are too large and affect accuracy sometimes. So in our experiments, K_d values are transformed into logspace (pK_d), same with [11], as

$$pK_d = -\log_{10} \left(\frac{K_d}{10^9} \right).$$

KIBA dataset The affinity of the KIBA dataset is obtained by combining the kinase inhibitor bioactivities from different sources such as K_d , K_i and IC_{50} through the method of KIBA [21]. The dataset contains 52498 drugs and 467 targets, with a total of 246088 KIBA scores. These scores are in the range of $[0, 17.2]$. We removed the drug–target pairs which have known affinities less than 10 in the KIBA dataset, and the final dataset we used contains 2111 drugs and 229 targets, with an affinity quantity density of 24.4%. The frequency histogram of the affinities in Davis and KIBA dataset is shown in Fig. 5.

The data of compounds and proteins in Davis dataset and KIBA dataset are strings, so the first step in our experiment is to convert strings to arrays. According to [11], drug SMILES with the length less than 85 and protein sequences with length less than 1200 in Davis dataset, and drug SMILES with the length less than 100 and protein sequences with length less than 1000 in KIBA dataset cover at least 90% of the compound SMILES strings and 80% of the protein sequences in each dataset, respectively. So in our experiment, we chose the above length as the maximum length in the training and test data. Because we need to encode characters into labels, so we extract all the different characters in all the strings and create a dictionary where the key is a

character and the value is an integer. In both datasets, drug SMILES strings have 64 different characters and target sequences have 25 different characters.

3.2 Evaluation Metrics

To evaluate the performance of our model, we used five evaluation metrics in our experiments to compare our method and the existing methods. To evaluate the Co-VAE's performance on generating new drugs, we use two metrics validity and uniqueness. Suppose the ground truth of the pairwise affinity for the test drug-target pairs is y , and the predicted affinity is $\hat{y}$.

Concordance index (CI) Concordance Index (CI) is defined as

$$CI = \frac{1}{Z} \sum_{\delta_i > \delta_j} \Phi(b_i - b_j),$$

where b_i is the prediction values for the larger true value δ_i , b_j is the prediction values for the smaller true value δ_j , Z is a normalization constant, $\Phi(x)$ is a piecewise function as follows:

$$\Phi(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0.5, & \text{if } x = 0 \\ 0, & \text{if } x < 0. \end{cases}$$

CI measures the probability that the predicted interactions are in the same order as the observed interactions, and its value ranges from 0 to 1. The closer the value is to 1, the better the model fits. CI is also used in [9] and [29].

Mean Squared Error (MSE) and Mean Absolute Error (MAE). MSE and MAE can measure the difference between the predicted affinities and the real affinities. MSE and MAE are defined as

$$MSE = \sum_i (y_i - \hat{y}_i)^2, MAE = \sum_i |y_i - \hat{y}_i|,$$

respectively.

r_m^2 index The r_m^2 index, which was also used in DeepDTA, can be used to evaluate the external predictive performance of quantitative structure–activity relationship (QSAR) models. It can be defined as

$$r_m^2 = r^2 * \left(1 - \sqrt{r^2 - r_0^2}\right),$$

where r^2 and r_0^2 are the squared correlation coefficients with and without intercept, respectively. Larger r_m^2 values for the test set indicate a more acceptable model.

Area Under Curve (AUC). The area under the ROC curve (AUC) could evaluate the performance of a classification model. The AUC can be considered as the probability that the classifier will rank a randomly chosen positive instance higher than a randomly chosen negative instance [31]. The range of AUC is 0 to 1, and a larger AUC value indicates better performance of a classifier. We only use AUC for the KIBA dataset, since the affinity distribution in Davis dataset is very uneven. To calculate AUC values, we take thresholds in the interval [10.5,12.5] with increment 0.1.

Validity and Uniqueness. The Co-VAE model can also generate new drugs for a given drug. To evaluate the model's

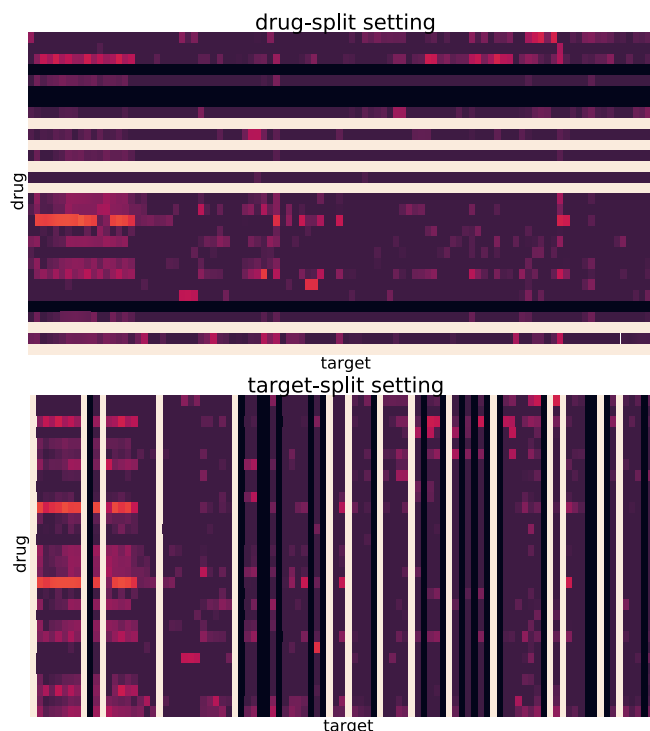


Fig. 6. The two experimental settings illustrated in affinities matrix from Davis dataset [20], where the rows and columns correspond to the drugs and targets, respectively. The affinities matrix is split into training data, test data (white part) and validation data (black part).

performance for generating drug sequences, we use two evaluation metrics, validity and uniqueness. Validity is defined as the proportion of valid drug SMILES sequences in all the generated drug SMILES sequences, where valid drug SMILES are determined through the tool Rdkit [30]. Uniqueness is defined as the proportion of new valid drugs among the overall valid drugs, where new valid drug means the valid drug SMILE sequences that never appear in the training data set.

3.3 Experimental Setting

To evaluate the performance of affinity prediction, we compare the Co-VAE method with DeepDTA [11], KronRLS [9], DeepAffinity [12] and GraphDTA [13] for two settings, a new-drug setting and a new-target setting. In the new-drug setting, drugs were randomly divided into six equal folds, five of which were considered as training drugs and the remaining fold was considered as the test drugs. Once the Co-VAE model is learnt on the training data, which consists of the training drugs and all the targets, it is used to predict the pairwise affinities between test drugs and all the targets. In the new-target setting, we randomly divided the targets to six folds. Similarly to the new-target setting, the Co-VAE model was learnt and used to predict the affinities between new targets and all the drugs. The two settings are shown in Fig. 6. For each setting in each dataset, we randomly split the drugs or targets for ten times, and the means and standard deviations for each evaluation metrics were reported. For a fair comparison, the same settings were used for all the comparison methods.

To perform the Co-VAE model and the DeepDTA model, several hyperparameters as shown in Table 1 should be

TABLE 1
The Parameter Settings of the Co-VAE Model

parameters	range
Number of filters in encoder	32*1;32*2;32*3
Number of filters in decoder	32*3;32*2;32*1
Filter length(drug SMILES strings)	[5,7]
Filter length(target sequence)	[7,11]
the rate of dropout	0.2
λ	[-3,-5]
epoch	100
batch size	256
learning rate	0.001

determined. The settings of the hyperparameters are almost the same for the two methods, except that the Co-VAE model has an extra regularization parameter λ . We fixed some of the hyperparameters in the experiments, including the number of filters in encoder, the number of filters in decoder and the rate of dropout, as shown in Table 1. We then determined the remaining hyperparameters by the strategy of five-fold cross-validation based on only training data. The filter size for drug SMILES strings was chosen from 5 and 7. For target sequences, which have longer strings than drug SMILES, we selected the corresponding filter size from the numbers 7 and 11. We choose -3 or -5 for the regularization parameter λ through cross-validation. DeepAffinity [12] and GraphDTA [13] were performed using the default parameters.

During the training process, we set the value of epoch to 100, the size of each batch to 256, and used the Adam [14] algorithm with learning rate of 0.001 to update the network parameters. In the experiment, for the Davis dataset, the drug SMILES strings and target protein sequences are embedded into matrices whose dimension are (85,128) and (1200,128). For the KIBA dataset, we convert drug SMILES

strings and target protein sequences into (100,128) and (1000,128) dimensional matrix. Our codes are all based on Pytorch and available on <https://github.com/LiminLi-xjtu/CoVAE>.

3.4 Experimental Results

We first evaluated the Co-VAE methods by plotting the ground truth and predicted affinities for the drug-target pairs. Figs. 7 and 8 showed the scatter plots of the ground truth affinities versus the predicted affinities for the Davis dataset and Kiba dataset, respectively. The results for new-drug setting and new-target settings were both reported in the two figures. An ideal method should generate a straight line $y = x$ (the blue line in the figures) in the scatter plots. Thus the more points are concentrated around the straight line, the better is the corresponding method. For the Davis dataset, in the top row of Fig. 7 with the new-drug setting, the points by the DeepDTA look the most scattered around the straight line, and those by the Co-VAE are the most concentrated around the line. For the GraphDTA, most points are concentrated below the straight line. In the bottom row of Fig. 7 for the new-target setting, the DeepDTA and DeepAffinity generated the most disperse points, KronRLS and GraphDTA generated some outlying points below the straight line, and the Co-VAE tend to generate the most concentrated and less outlying points. For the KIBA dataset, as shown in Fig. 8, KronRLS generated the most scattered points that are mostly below the straight line, DeepDTA, DeepAffinity and GraphDTA tend to generate more outlying points, and the Co-VAE appears to generate the most concentrated points. In summary, by the scatter plots of the two figures, we could intuitively compare the performance of the five methods, and our Co-VAE performed the best overall.

We then evaluated the performance of Co-VAE method on drug-target affinity prediction by four quantitative

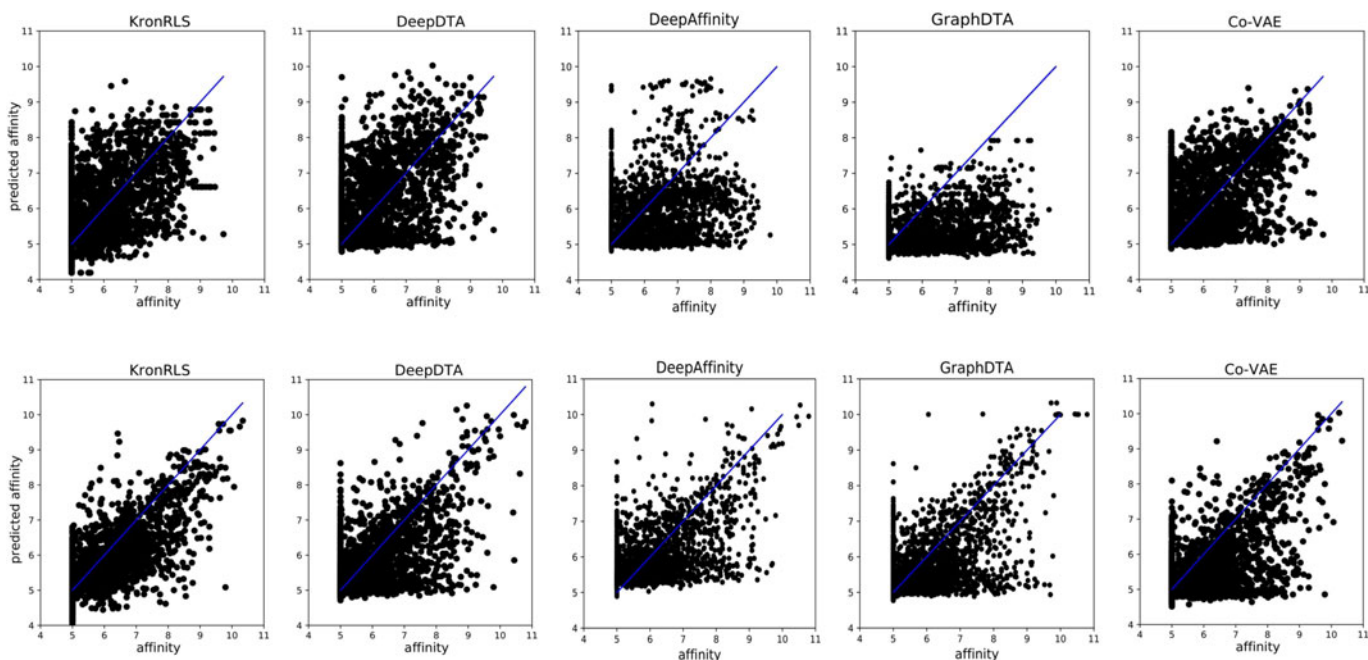


Fig. 7. Ground truth affinities (x -axis) versus predicted affinities (y -axis) for new-drug setting (top) and new-target setting (bottom) by five methods on the Davis dataset.

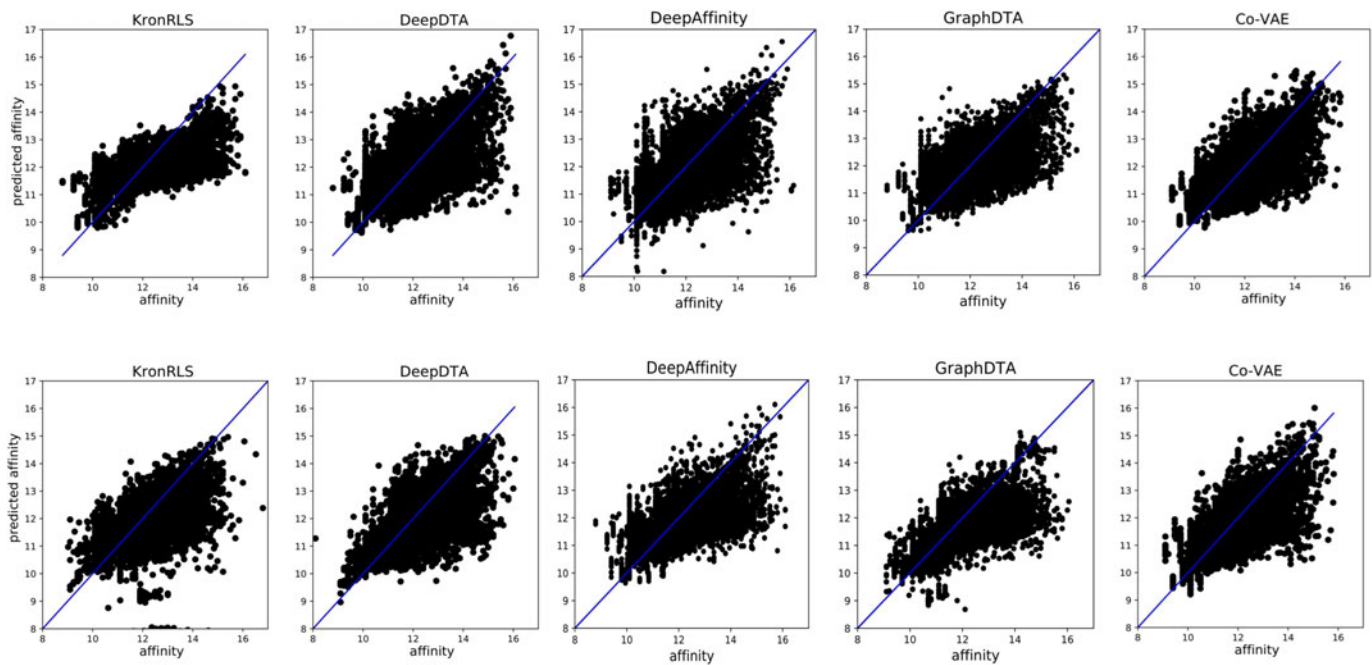


Fig. 8. Ground truth affinities(x -axis) versus predicted affinities (y -axis) for new-drug setting (top) and new-target setting (bottom) by five methods on the KIBA dataset.

TABLE 2 The Average CI, MSE, MAE and r_m^2 Obtained by Five Methods for the Davis Dataset and KIBA Dataset With Two Settings						
Dataset	Settings	Method	CI(std)	MSE(std)	MAE(std)	r_m^2
Davis	New-drug	KronRLS	0.656 (0.033)	0.796 (0.231)	0.606 (0.086)	0.143
		DeepDTA	0.671 (0.023)	0.726 (0.091)	0.527(0.033)	0.122
		DeepAffinity	0.673 (0.057)	0.955 (0.140)	0.639 (0.040)	0.114
		GraphDTA	0.732 (0.067)	0.840 (0.058)	0.570 (0.020)	0.176
		Co-VAE	0.712 (0.063)	0.724 (0.096)	0.550 (0.048)	0.107
	New-target	KronRLS	0.836 (0.014)	0.429 (0.055)	0.428 (0.025)	0.448
		DeepDTA	0.780 (0.071)	0.490 (0.095)	0.456 (0.047)	0.433
		DeepAffinity	0.803 (0.006)	0.477 (0.019)	0.434 (0.050)	0.444
		GraphDTA	0.778 (0.016)	0.457 (0.009)	0.397 (0.011)	0.429
		Co-VAE	0.816 (0.011)	0.419 (0.016)	0.380 (0.013)	0.477
KIBA	New-drug	KronRLS	0.729 (0.005)	0.449 (0.005)	0.457 (0.003)	0.355
		DeepDTA	0.728 (0.007)	0.466 (0.022)	0.419 (0.011)	0.342
		DeepAffinity	0.707 (0.016)	0.526 (0.024)	0.491 (0.023)	0.283
		GraphDTA	0.728 (0.008)	0.442 (0.012)	0.447 (0.019)	0.379
		Co-VAE	0.742 (0.011)	0.416 (0.004)	0.401 (0.007)	0.358
	New-target	KronRLS	0.591 (0.025)	0.825 (0.042)	0.603 (0.010)	0.399
		DeepDTA	0.722 (0.017)	0.430 (0.016)	0.421 (0.013)	0.114
		DeepAffinity	0.700 (0.007)	0.490 (0.016)	0.470 (0.011)	0.324
		GraphDTA	0.658 (0.050)	0.519 (0.045)	0.504 (0.027)	0.314
		Co-VAE	0.741 (0.010)	0.421 (0.010)	0.396 (0.005)	0.356

metrics including CI, MSE, MAE and r_m^2 . Table 2 shows the means and standard deviations of the four evaluation metrics by the five compared methods on the Davis dataset. For the new-drug setting, the Co-VAE obtained the best MSE and second better CI and MAE. DeepDTA obtained a little bit better MAE than the Co-VAE, and GraphDTA obtained a little bit better CI (0.02) than the Co-VAE, but the Co-VAE obtained much better MSE than the Co-VAE, and much better CI than DeepDTA. All the three methods could not obtain good r_m^2 in this setting. To further evaluate the

performance of the Co-VAE on each single drug, we reported the MSE for each test drug in this dataset, and the results were shown in Fig. 9. The figure shows that the Co-VAE could obtain the best affinities or slightly worse affinities than the best prediction for almost all the drugs, for example, GraphDTA obtained slightly better MSE for drug with Pubchem CID 11617559 and 6450551. For the new-target setting, the Co-VAE method performs the best by three of the four metrics(MSE, MAE and r_m^2), and it obtains a slightly worse CI than KronRLS. For the KIBA dataset, the

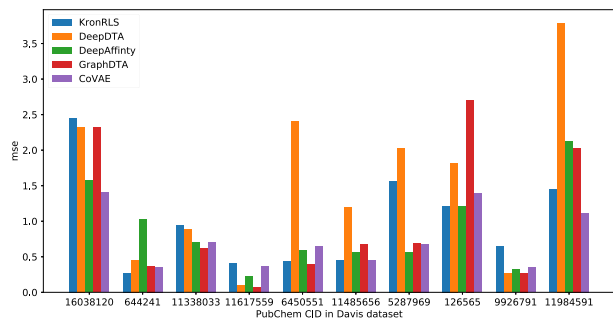


Fig. 9. MSE for each single test drug in Davis dataset by five different methods.

means and standard deviations of the four evaluation metrics by the three compared methods were reported in Table 2. For both new-drug setting and new-target setting, the Co-VAE method performs the best for three metrics (CI, MSE and MAE) and the second best for the other metric (r_m^2). We also found that for the new-target setting in the KIBA dataset, the KronRLS method performs significantly worse than the two deep learning methods.

We further evaluated the Co-VAE for its performance on drug-target interaction prediction by the evaluation metric AUC. By fixing a cutoff value of the ground truth affinity value, the drug-target pairs were labelled as positive and negative. AUC values were then calculated with the labels and the predicted affinities. Fig. 10 shows the AUC values by using different cutoff values in (10.5,12.5) for the KIBA dataset in two settings. We can see from the figure that our method Co-VAE performs stably better than the other four methods for both new-drug setting and new-target setting, except DeepDTA performs a little better than the Co-VAE for cutoff value larger than 11.75 for new-drug setting, and KronRLS performs a little better than the Co-VAE for a small range of cutoff values between 11.5 and 11.75.

We also applied the Co-VAE model to find out the drugs that have high affinities with the SARS-CoV-2 genes. We first downloaded the protein sequences for SARS-CoV-2 genes from NCBI,¹ and then applied the pretrained Co-VAE model on KIBA dataset to predict the binding affinities between these proteins and all the drugs in KIBA dataset. For each SARS-CoV-2 gene, we reported the top 5 drugs that are predicted to have highest affinities with the protein sequence. Due to the lack of space, we put the prediction results in <https://github.com/LiminLi-xjtu/CoVAE>. The predicted top drugs all have affinities larger than 12, which implies that the drugs are highly likely to have interactions with the corresponding proteins.

Finally, we applied the trained Co-VAE model by the KIBA dataset to generate new drugs for input drugs, which are expected to have similar affinity property with the input drug. We evaluated the generative performance of the Co-VAE model by two measurements, including validity, which is the percentage of the valid drug SMILES through Rdkit [30], and uniqueness, which is the percentage of new valid drugs which are not in the KIBA dataset. We compared the Co-VAE model with VAE, AAE[33], GAN[32]

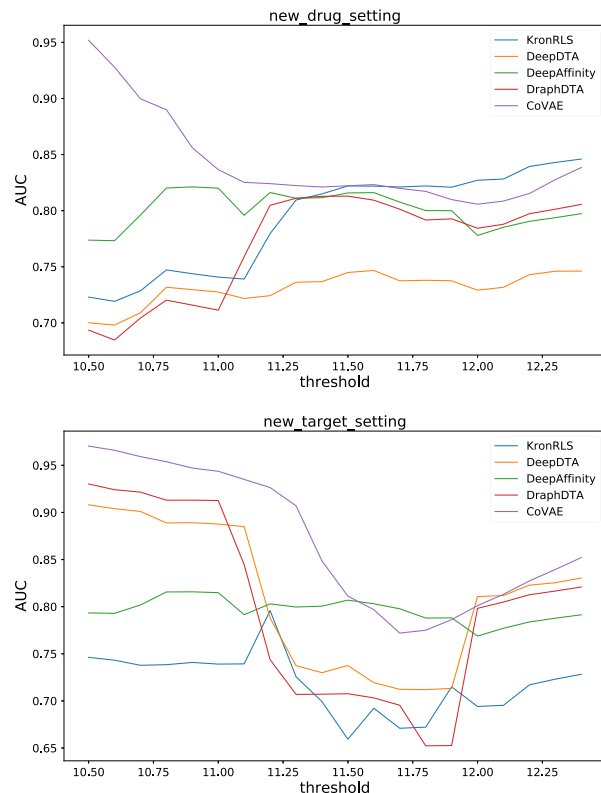


Fig. 10. The AUC curves obtained by five methods on KIBA dataset with new-drug setting and new-target setting.

and seqGAN[34]. The results of the newly generated drugs are shown in Table 3. In our experiments GAN and seqGAN could not generate effective drugs possibly due to the small training data. The Co-VAE could obtain the highest validity and uniqueness among all the models. This implies that our Co-VAE model could generate more effective new drugs than the comparison methods. We note that these methods are all less effective than other computational methods in drug discovery such as REINVENT [35] and ORGAN [36]). The reason might be that the Co-VAE and the comparison methods only use a relatively small dataset including drug and target structures, while other computational methods such as REINVENT or ORGAN use much larger datasets which include structures and other characteristics like solubility (a property that how likely a molecule is able to mix with water). This implies that the generative function of the Co-VAE model has space to be improved by involving other drug characteristics.

We finally give some examples of the generated drug SMILES in Tables 4 and 5. Table 4 shows some examples

TABLE 3
The Statistical Analysis of the Generated Drugs by GAN, SeqGAN, AAE, VAE and Co-VAE on the KIBA Dataset

model	Validity(%)	Uniqueness(%)
GAN	< 1	-
SeqGAN	< 1	-
AAE	1.26	0.69
VAE	48.54	3.25
Co-VAE	54.53	3.47

1. <https://www.ncbi.nlm.nih.gov/datasets/coronavirus/proteins/>

TABLE 4
New Generated Drugs That Could be Found in KIBA Dataset, With the Corresponding Input Drugs and Their PubChem CID, SMILES Strings and Activated Target Names

	PubChem CID	Drug SMILES strings (based on PubChem)	Activated target
Input drug	49830947	<chem>C1=CC=C2C(=C1)NC3=NN=C(C=C3C(=O)N2)Cl</chem>	1.AURKA - aurora kinase A (human) 2. CDK1 - cyclin dependent kinase 1 (human) 3.FLT4 - fms related tyrosine kinase 4 (human) 4.CHEK2 - checkpoint kinase 2 (human) ...
Generated drug	11993789	<chem>C1=CC=C2C(=C1)NC3=NN=C(C=C3C(=O)N2)I</chem>	1.AURKA - aurora kinase A (human) 2. CDK1 - cyclin dependent kinase 1 (human) 3.FLT4 - fms related tyrosine kinase 4 (human) 4.HIPK2 - homeodomain interacting protein kinase 2 (human) ...
Input drug	49830327	<chem>CCCC(=O)NC1=NC(=CC(=C1)C2=CNC3=C2C=CC=N3)Cl</chem>	1.CDC7 - cell division cycle 7 (human) 2.CLK4 - CDC like kinase 4 (human) 3.ROCK2 - Rho associated coiled-coil containing protein kinase 2 (human) 4.FLT1 - fms related tyrosine kinase 1 (human) ...
Generated drug	49830337	<chem>CCCCC(=O)NC1=NC(=CC(=C1)C2=CNC3=C2C=CC=N3)Cl</chem>	1.CDC7 - cell division cycle 7 (human) 2.CLK4 - CDC like kinase 4 (human) 3.ROCK2 - Rho associated coiled-coil containing protein kinase 2 (human) 4.CSNK1A1 - casein kinase 1 alpha 1 (human) ...
Input drug	49830636	<chem>CN1CCN(CC1)CC2=CC3=C(C=C2)C4=C(C3)C(=NN4)C5=CSC(=C5)C#CCOCC6CC6</chem>	1.AURKB - aurora kinase B (human) 2.BLK - BLK proto-oncogene, Src family tyrosine kinase (human) 3.CAMK2B - calcium/calmodulin dependent protein kinase II beta (human) 4.GSK3B - glycogen synthase kinase 3 beta (human) ...
Generated drug	16220223	<chem>CN1CCN(CC1)CC2=CC3=C(C=C2)C4=C(C3=O)C(=NN4)C5=CSC(=C5)C#CCOCCOC</chem>	1.AURKB - aurora kinase B (human) 2.BLK - BLK proto-oncogene, Src family tyrosine kinase (human) 3.CAMK2B - calcium/calmodulin dependent protein kinase II beta (human) 4.STK17A - serine/threonine kinase 17a (human) ...

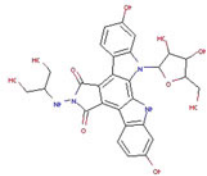
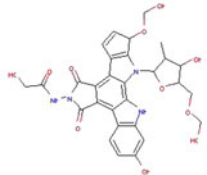
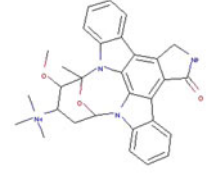
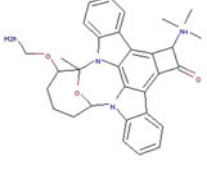
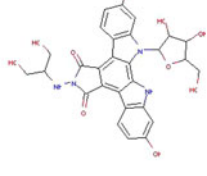
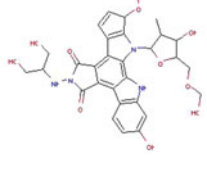
of the generated SMILES strings which can be found in the KIBA datasets, for which we also report the corresponding input drug SMILES, and check their IDs and targets from the database PubChem.² From this table, we can see that the generated drug SMILES strings are often very similar to the original input drug SMILES strings, with only a small part changed. For example, when we take the drug SMILES with ID 49830947 as the input, the Co-VAE model will generate a new SMILES with ID 11993789, which only changes the last element of the input SMILES. Furthermore, as expected, the Co-VAE could generate new drugs which share similar targets as the input drugs, as shown in the last column of the table. Table 5 shows some examples of the generated SMILES strings which can not be found in the KIBA datasets. We also report the corresponding input drug SMILES and their chemical structures drawn by Rdkit [30]. Since the generated SMILES could not be found in PubChem, their targets are unknown. However, we could see that the generated SMILES strings and the corresponding chemical structure diagram are both similar with those of the input drugs, respectively.

4 CONCLUSION

In our work, we propose a novel Co-VAE model with deep neural networks (GatedCNN) to predict drug-target binding affinity based on drug structures and target sequences. The objective function of the Co-VAE model consists of a drug VAE term (drug encoder and drug decoder) for generating drug SMILES strings, a target VAE term (target encoder and target decoder) for generating target sequences, and a co-regularized term (Reg block) which generates the drug-target binding affinity. Thus the Co-VAE model has two functions: predicting drug-target binding affinity and generating new drug SMILES which share similar targets with the input drugs (or target sequences which share similar drugs with the input targets). We theoretically explain the Co-VAE model and evaluated the two functions of the Co-VAE model by applying it to two drug-target binding affinity datasets. The results show that our proposed Co-VAE model could predict drug-target binding affinity for new drugs or new targets effectively with better performance than the existing models KronRLS, DeepDTA, DeepAffinity and GraphDTA in most cases. The generative function of the Co-VAE model is also evaluated by comparing it with GAN, VAE, AAE and SeqGAN.

2. <https://pubchem.ncbi.nlm.nih.gov/>

TABLE 5
New Generated Drugs That are not in KIBA Dataset, With the Corresponding Input Drugs
and Their SMILES Strings and Chemical Structure Diagram Draw by Rdkit

	Drug SMILES strings	Chemical structure diagram
Input drug	<chem>C1=CC2=C(C=C1O)NC3=C4C(=C5C(=C23)C(=O)N(C5=O)NC(CO)CO)C6=C(N4C7C(C(C(O7)CO)O)O)C=C(C=C6)O</chem>	
Generated drug	<chem>C1=CC2=C(C=C1O)NC3=C4C(=C5C(=C23)C(=O)N(C5=O)NC(=O)CO)C6=C(N4C7C(C(C(O7)COCO)O)C)C(C=C6)OCO</chem>	
Input drug	<chem>CC12C(C(CC(O1)N3C4=CC=CC=C4C5=C6C(=C7C8=C(C=CC=C8N2C7=C53)CNC6=O)[N+](C)(C)C)OC</chem>	
Generated drug	<chem>CC12C(CCCC(O1)N3C4=CC=CC=C4C5=C6C(=C7C8=CC=CC=C8N2C7=C53)C(C6=O)[N+](C)(C)C)OCN</chem>	
Input drug	<chem>C1=CC2=C(C=C1O)NC3=C4C(=C5C(=C23)C(=O)N(C5=O)NC(CO)CO)C6=C(N4C7C(C(C(O7)CO)O)O)C=C(C=C6)O</chem>	
Generated drug	<chem>C1=CC2=C(C=C1O)NC3=C4C(=C5C(=C23)C(=O)N(C5=O)NC(CO)CO)C6=C(N4C7C(C(C(O7)COCO)O)C)C(C=C6)OCO</chem>	

Although the Co-VAE model is designed for predicting drug-target binding affinity, it has many potential applications such as in image processing or natural language processing, for example, to match the images and text captions, match the documents to abstracts, and so on. One further research direction is to explore the potential applications of the Co-VAE in more general applications. Another further research

is to improve the generative performance of the Co-VAE model by involving other data sets such as drug solubility.

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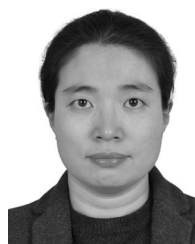
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Tianjiao Li received the BS degree from the School of Mathematics, Jilin University, in 2018. He is currently working toward the graduate program for the MS degree with the school of mathematics and statistics, Xi'an Jiaotong University, Xi'an, China. His research interest include machine learning.



Xing-Ming Zhao (Senior Member, IEEE) received the PhD degree from the University of Science and Technology of China, Hefei, China in 2005. He is currently a full professor at the Institute of Science and Technology for Brain Inspired Intelligence, Fudan University, China, and serves as a Vice Director of the Key Laboratory of Computational Neuroscience and Brain-Inspired Intelligence (Fudan University), Ministry of Education, China. His research interests include data mining, bioinformatics, and computational systems biology. He has published more than 100 peer-reviewed journal papers. He is a Co-Chair of IEEE SMC Technical Committee on Systems Biology. He is also the lead guest editor and the editorial member of several journals, e.g. IEEE/ACM TCBB, Neurocomputing, Journal of Theoretical Biology, IET Systems Biology, and so on.



Limin Li received the bachelor's and master's degrees from Zhejiang University in 2004 and 2006, respectively and the PhD degree in mathematics from the University of Hong Kong in 2010. She was a postdoctoral fellow with Max Planck Institute of Intelligent System. She is currently a professor with the school of mathematics and statistics in Xi'an Jiaotong University, Xi'an, China. Her research interests include machine learning and the applications in bioinformatics.

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